Sample of working initialization for a RISS GPS/INS system in loosely coupled mode.

Odometer and vertical gyroscope parameters for KVH and VTI sensors.

```
% KVH
bwz.corr_time = 3*3600;%3 hours (gyro correlation time)
bwz.stdv      = 0.01*pi/180;% (gyro standard deviation)

a_od.corr_time = 0.1*3600; % (odom correlation time)
a_od.stdv      = 10;       % (odom standard deviation)

% VTI
bwz.corr_time = 4*3600; %  4 hours
bwz.stdv      =0.1*pi/180;

a_od.corr_time = .1*3600;
a_od.stdv      = 10;
```

Assuming a 9 states Kalman filter:

Below are the numbers for the KVH data but VTI have close numbers as well.

The noise coupling matrix (G)

```
G = diag(ones(1,9));
G(8,8) = sqrt(2*(1/a_od.corr_time)*a_od.stdv^2);
G(9,9) = sqrt(2*(1/bwz.corr_time)*bwz.stdv^2);
For example, in the case of VTI sensor, given the above
numbers G would look like:
G =
1.0    0     0     0     0     0     0      0      0
 0    1.0    0     0     0     0     0      0      0
 0     0    1.0    0     0     0     0      0      0
 0     0     0    1.0    0     0     0      0      0
 0     0     0     0    1.0    0     0      0      0
 0     0     0     0     0    1.0    0      0      0
 0     0     0     0     0     0    1.0     0      0
 0     0     0     0     0     0     0    0.7454   0
 0     0     0     0     0     0     0      0   2.0569e-05
```

The system covariance matrix (Q)

```
Q(9,9)=0;
Q(1,1) = 1e-15;  % lat
Q(2,2) = 1e-15; %long
Q(3,3) = 1e-15; % h
Q(4,4) = 0; % ve (already initialized but just to show you
Q(5,5) = 0; % vn which quantity belongs to what variable )
Q(6,6) = 0; % vu
Q(7,7) = 0; % A
Q(8,8) = 1e-5; % a_od
Q(9,9) = 0.001*pi/180 ; % Bz

Qd = G*Q*G'*dt; % then use Qd
```

The initial covariance matrix (P)

```
P=eye(9); % 9*9 identity matrix

P(1,1) = 0.00157; % lat
P(2,2) = 0.00156; %long
P(3,3) = 100000; % h
P(4,4) = 160*10000/3600; % ve
P(5,5) = 160*10000/3600; % vn
P(6,6) = 160*10000/3600; % vu
P(7,7) = pi ; % Azimuth (pi the constant 3.1416)
P(8,8) = 100; % a_od
P(9,9) = 10; %Bz
```

The measurement covariance matrix (R)

The measurement covariance matrix is initialized based on the information provided by the GPS reviver but here is how. In the below calculations: k is the current epoch index, M (meridian radius of curvature) is calculated each time as given in the course notes. Make sure that latitude and longitude variables are in radians when using them in your calculation.

```
lat_std_in_meters = Lat_std(k); % given in data set
long_std_in_meters = Long_std(k); % given in data set

lat_std_in_rad = lat_std_in_meters/(M+Alt(k));
long_std_in_rad = long_std_in_meters/((M+Alt(k))*cos(Lat(k)));
```

```
R(1,1) = (lat_std_in_rad)^2;
R(2,2) = (long_std_in_rad)^2;
R(3,3) = (Alt_std(k))^2;
R(4,4) = (long_std_in_meters*5)^2;
R(5,5) = (lat_std_in_meters*5)^2;
R(6,6) = (Alt_std(k)*10)^2;
```

One more step: in real implementation, numbers in many cases do not make sense and hence your Kalman filter will never converge. Thus, you need to do some empirical tuning. We often scale R by a scale factor:

R = R * ScaleFactor ;

For the NovAtel GPS receiver data ScaleFactor= 0.0045 works well. So, after you initialize R from the variances given by the receiver each epoch, use:

```
R = R * .00045;
```

Remember that R and Q are somehow proportional so, thus scale factor works well with the Q as initialized above but might not work if you play with the numbers in Q. if you did, then you might need to change this scale factor accordingly.